Problem Set 10

Problem 6.1

(a) Starting with the definition of the Helmholtz energy in Equation 6.3, derive the fundamental property relation in Equation 6.10

(b) Starting with the definition of the Gibbs energy in Equation 6.4, derive the fundamental property relation in Equation 6.11.

Problem 6.4

(a) Starting with the fundamental property relation Equation 6.10, derive the Maxwell relation given in Equation 6.16.

(b) Starting with the fundamental property relation Equation 6.9, derive the Maxwell relation given in Equation 6.15.

Problem 6.141

Calculate Z, H^R, and S^R by the Redlich-Kwong equation for the following:

(a) Ethylene at 300 K and 35 bar.
(b) Hydrogen sulfide at 400 K and 70 bar.
(c) Nitrogen at 150 K and 50 bar.
(d) n-Octane at 575 K and 15 bar.
(e) Propane at 375 K and 25 bar.